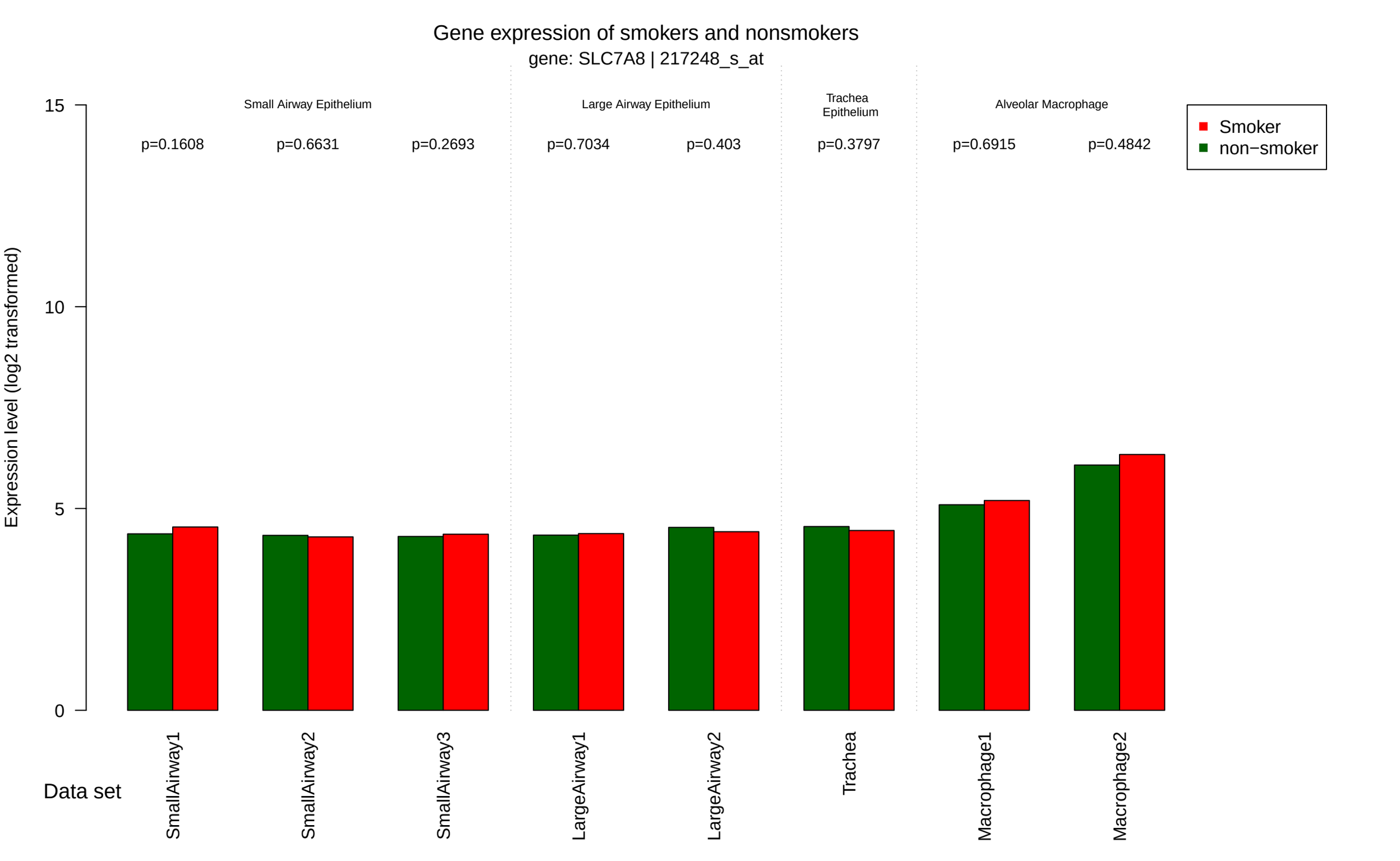
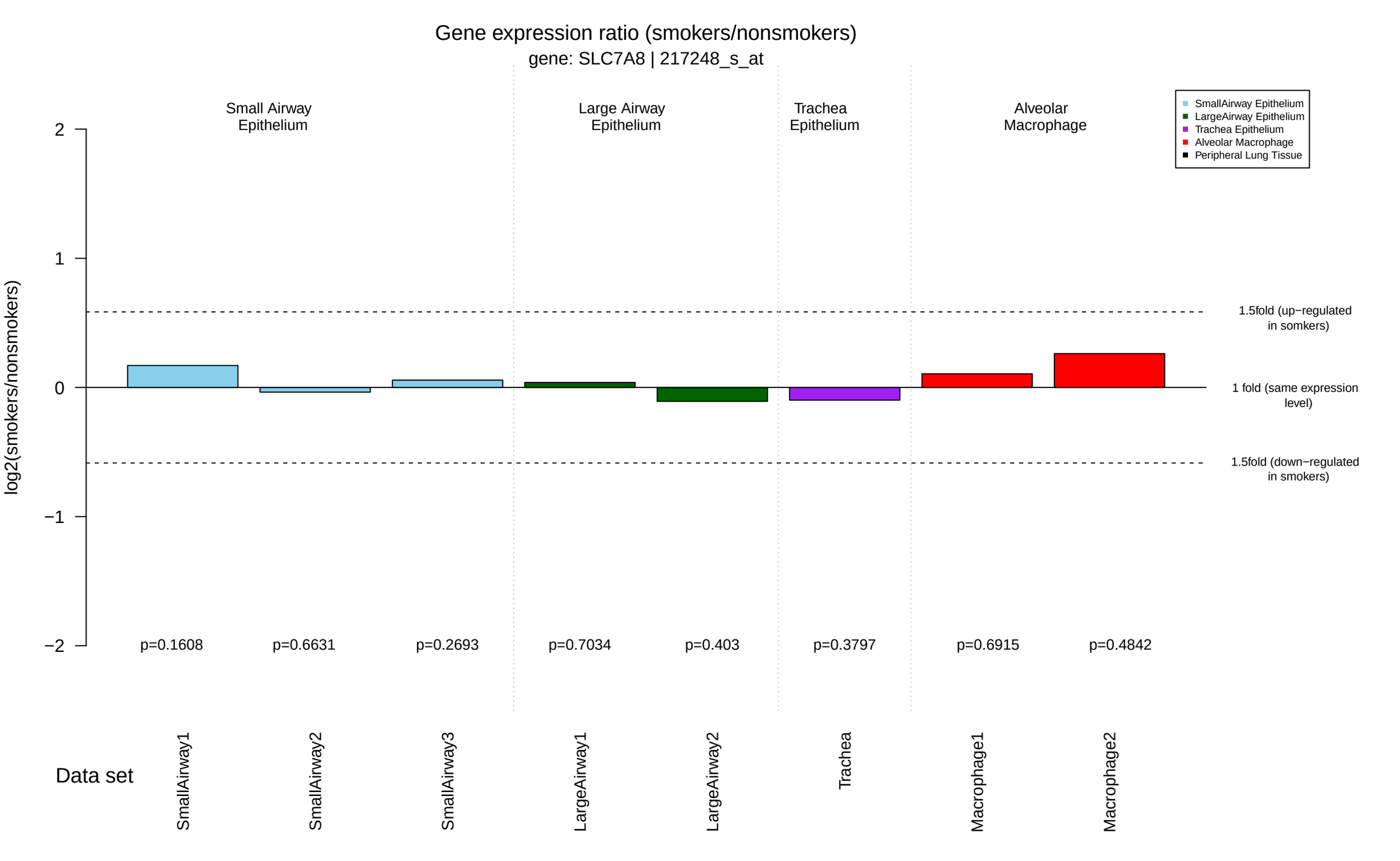
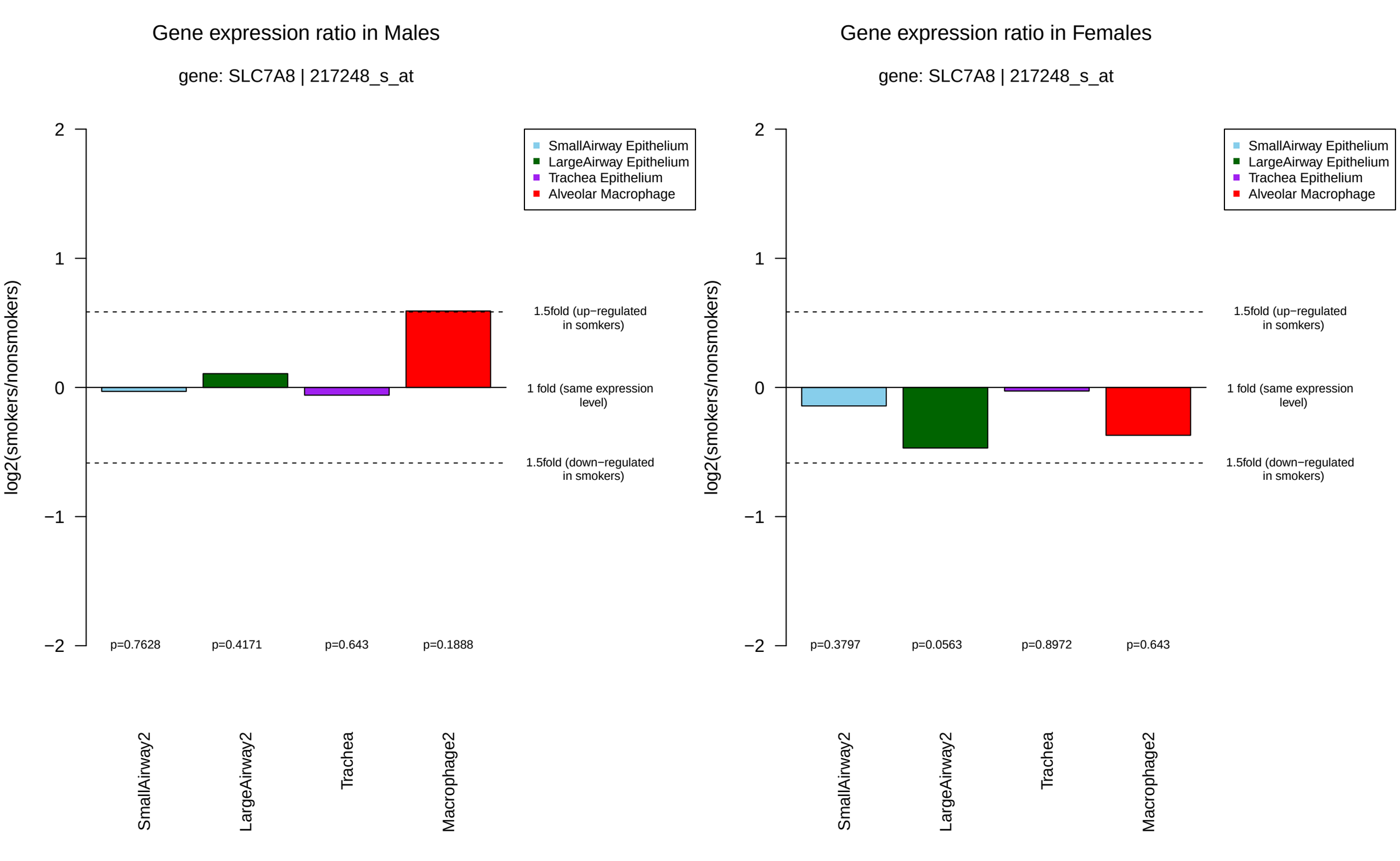
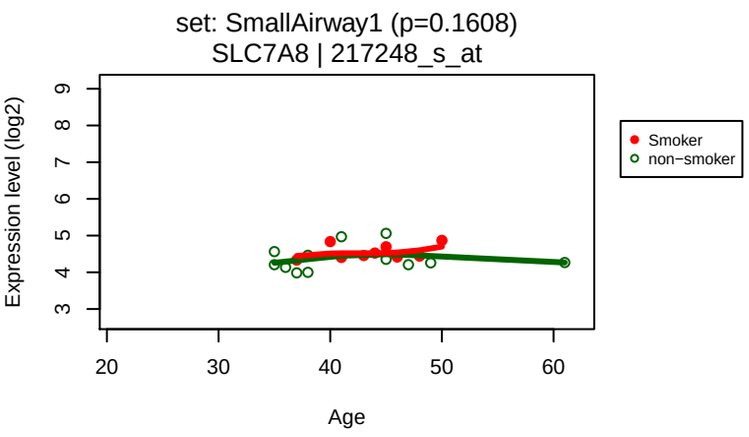
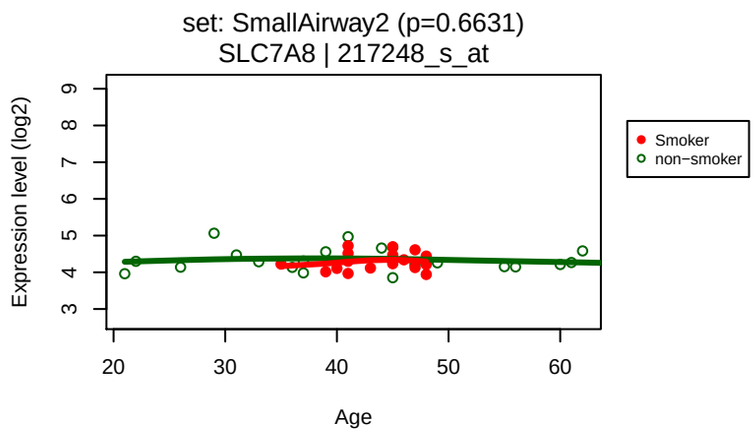
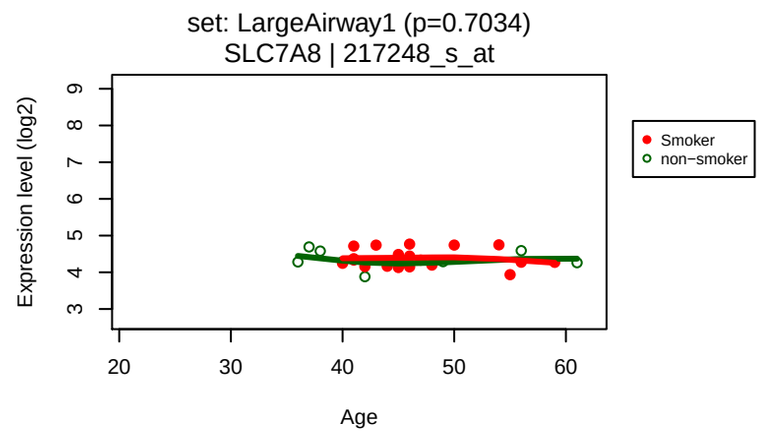

Gene expression of smokers and nonsmokers
gene: SLC7A8 | 217248_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1608
p=0.6631
p=0.2693
p=0.7034
p=0.403
p=0.3797
p=0.6915
p=0.4842
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

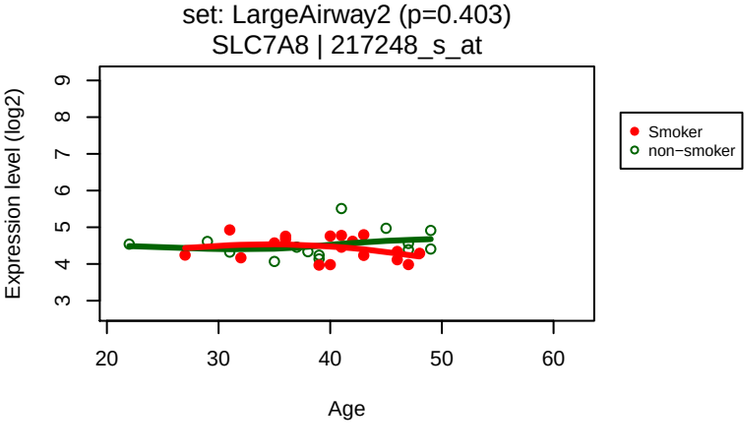

Gene expression ratio (smokers/nonsmokers)
gene: SLC7A8 | 217248_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1608
p=0.6631
p=0.2693
p=0.7034
p=0.403
p=0.3797
p=0.6915
p=0.4842
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

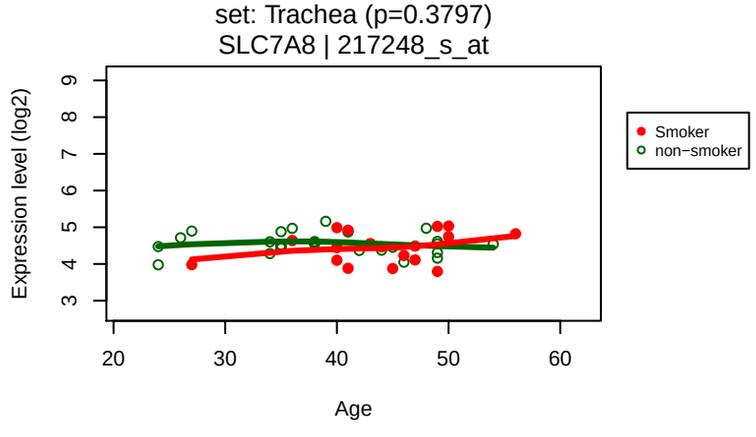

Gene expression ratio in Males
gene: SLC7A8 | 217248_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7628
p=0.4171
p=0.643
p=0.1888
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC7A8 | 217248_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3797
p=0.0563
p=0.8972
p=0.643
SmallAirway2
LargeAirway2
Trachea
Macrophage2

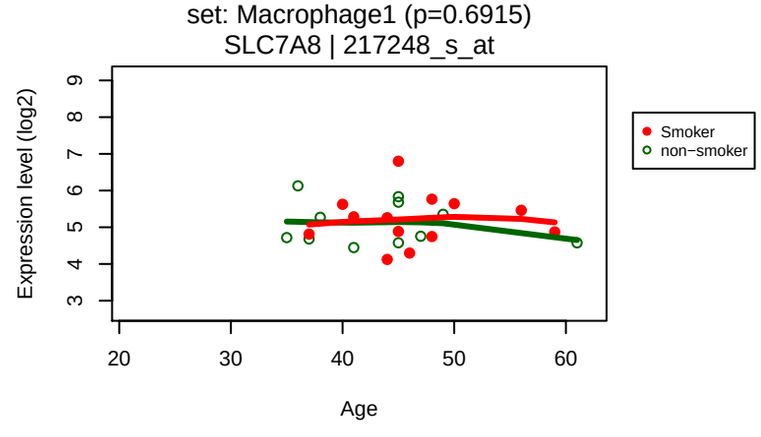

set: SmallAirway1 (p=0.1608)
SLC7A8 | 217248_s_at
Expression level (log2)
Age
Smoker
non−smoker

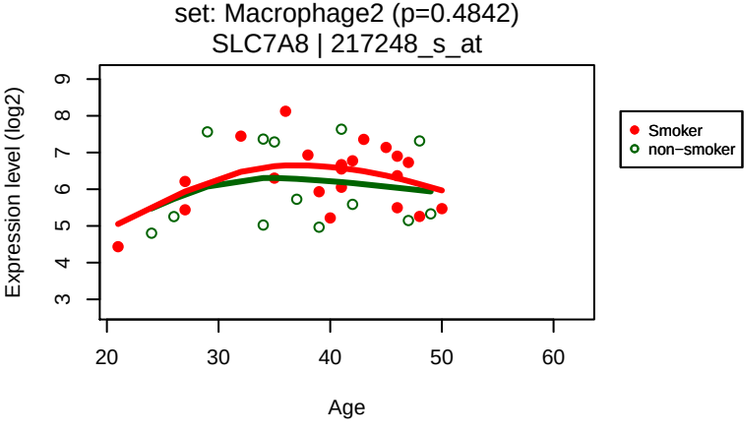

set: SmallAirway2 (p=0.6631)
SLC7A8 | 217248_s_at
Expression level (log2)
Age
Smoker
non−smoker

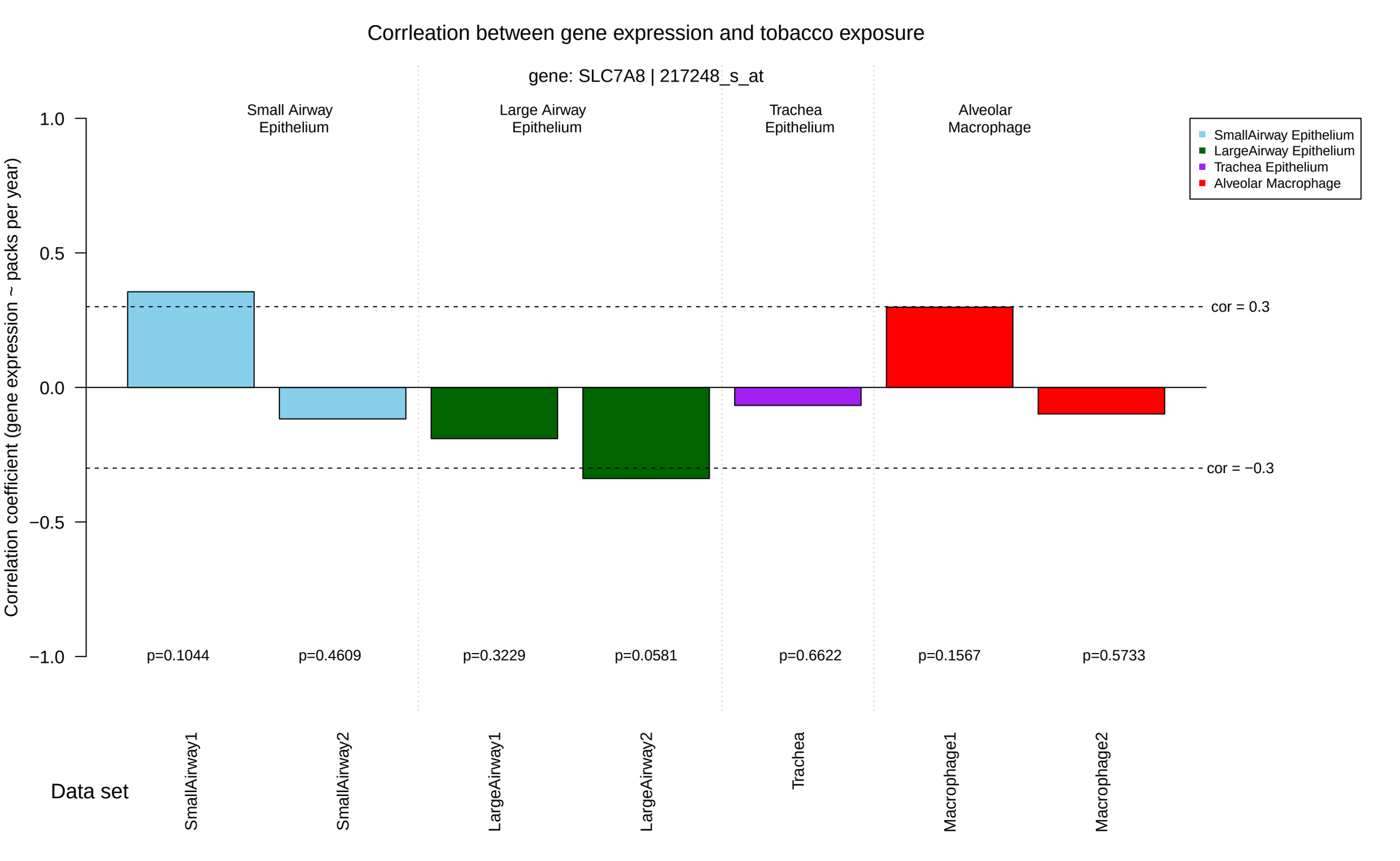

set: LargeAirway1 (p=0.7034)
SLC7A8 | 217248_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.403)
SLC7A8 | 217248_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3797)
SLC7A8 | 217248_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6915)
SLC7A8 | 217248_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4842)
SLC7A8 | 217248_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC7A8 | 217248_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1044
p=0.4609
p=0.3229
p=0.0581
p=0.6622
p=0.1567
p=0.5733
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2